Harsh Kumar

+1-416-919-8537 | harsh@cs.toronto.edu | LinkedIn | Google Scholar| harsh-kumar.com

Education

University of Toronto — Toronto, ON
Ph.D. in Computer Science — *September 2021 – Present*
Supervisor: Dr. Ashton Anderson

- Relevant courses: Introduction to Machine Learning, Designing Intelligent Self-Improving Systems, Ubiquitous Computing and Mobile Health, Computational Social Science, Software Engineering for ML Systems, AI Alignment.

Vellore Institute of Technology — Tamil Nadu, India
B.Tech. in Computer Science and Engineering — *July 2016 – July 2020*
CGPA: 8.99/10

- Capstone project: Developed a chatbot framework for the domain of investment banking, as part of a 6-month internship at JPMorgan Chase.

Employment

Graduate Research Assistant — September 2021 – Present
University of Toronto — *Toronto, ON*

- Research on AI + society with emphasis on rigorous measurement: designing algorithms, studies, and evaluation frameworks to quantify how AI systems affect cognition, well-being, and learning.
- Led large-scale human-centered experiments, such as a randomized controlled trial on LLM-assisted math learning, to identify mechanisms and boundary conditions for beneficial impact.
- Conducted controlled studies on how repeated LLM use shapes creative ideation and convergence, informing how post-deployment dynamics can shift human behavior over time.
- Designing LLM agents for coaching and social support; earlier built contextual-bandit methods to personalize mental health support to individuals' evolving needs.

Graduate Teaching Assistant — September 2021 – Present
University of Toronto — *Toronto, ON*

- TA for CSC428 (HCI), CSC2558 (Multidisciplinary HCI), CSC2552 (Computational Social Science), CSC318 (Design of Interactive Media), CSC108 (Intro to Programming); supported student research/assignments emphasizing experimental design, measurement, and communication of findings to mixed audiences.

Research Intern — May 2025 – August 2025
Microsoft, Viva Insights — *Redmond, USA*

- Applied Science research internship under Dr. Shamsi Iqbal: designed LLM-based agents and evaluation approaches to help business leaders understand AI adoption, map organizational risks/opportunities, and take actions that support sustainable deployments. Collaborated with cross-functional partners to translate findings into decision-relevant product directions.

Research Intern — May 2024 – August 2024, May 2023 – August 2023
Microsoft Research — *New York City, USA*

- Interned at Microsoft Research (Computational Social Science): designed controlled experiments and evaluation pipelines to study how LLMs augment human cognition, with a focus on AI for Education (e.g., GPT-4, Phi-3) and measurable learning outcomes.

Software Engineer — January 2020 – August 2021
J.P.Morgan Chase & Co. — *Mumbai, India*

- Built an end-to-end NLU system using an LSTM model with user-feedback-driven improvement loops, spanning UI design, model training/deployment, and data pipelines. Worked with production constraints (scalability, reliability, CI/CD), building practical infrastructure for evaluating and iterating on model behavior in real-world settings.
- *Skills:* Python - Django, Flask, Tensorflow, PyTorch. Tech stacks - LAMP, MERN. CI/CD.

Research Analyst Intern — May 2019 – January 2020
McKinsey & Co. — *Gurugram, India*

- Contributed to a semi-autonomous UAV for warehouse operations; developed the vision system for object detection and avoidance using visual SLAM, focusing on perception and evaluation under operational constraints.
- *Skills:* Pytorch, OpenCV, NumPy, Microcontrollers.

Accepted Publications

- **Harsh Kumar**, Jasmine Chahal, Yinuo Zhao, Zeling Zhang, Annika Wei, Louis Tay, Ashton Anderson
  When AI Gives Advice: Evaluating AI and Human Responses to Online Advice-Seeking for Well-Being
  (Just accepted) ACM CHI Conference on Human Factors in Computing Systems (CHI), 2026 arXiv:2512.08937
- Jiayin Zhi, **Harsh Kumar**, Mina Lee
  Investigating the Effects of LLM Use on Critical Thinking Under Time Constraints
  (Just accepted) ACM CHI Conference on Human Factors in Computing Systems (CHI), 2026
- **Harsh Kumar**, Suhyeon Yoo, Angela Z. Bernuy, Jiakai Shi, Huayin Luo, Joseph Williams, Anastasia Kuzminykh, Ashton Anderson, Rachel Kornfield
  Large Language Model Agents for Improving Engagement with Behavior Change Interventions: Application to Digital Mindfulness
  Proceedings of the ACM on Human-Computer Interaction (PACMHCI), CSCW, 2025 doi:10.1145/3757619
- Ananya Bhattacharjee, Joseph Jay Williams, Miranda Beltzer, Jonah Meyerhoff, **Harsh Kumar**, Haochen Song, David C. Mohr, Alex Mariakakis, Rachel Kornfield
  Investigating the Role of Situational Disruptors in Engagement with Digital Mental Health Tools **(Honorable Mention Award, Top 5% of Submissions)**
  Proceedings of the ACM on Human-Computer Interaction (PACMHCI), CSCW, 2025 doi:10.1145/3757487
- **Harsh Kumar**, David M. Rothschild, Daniel G. Goldstein, Jake M. Hofman
  Math Education With Large Language Models: Peril or Promise?
  International Conference on Artificial Intelligence in Education (AIED), 2025
  link.springer.com/chapter/10.1007/978-3-031-98459-4_5
- **Harsh Kumar**, Jonathan Vincentius, Ewan Jordan, Ashton Anderson
  Human Creativity in the Age of LLMs: Randomized Experiments on Divergent and Convergent Thinking
  (Honorable Mention Award, Top 5% of Submissions)
  ACM CHI Conference on Human Factors in Computing Systems (CHI), 2025 doi:10.1145/3706598.3714198
- Ilya Musabirov, Mohi Reza, Haochen Song, Steven Moore, Pan Chen, **Harsh Kumar**, Tong Li, John Stamper, Norman Bier, Anna Rafferty, Thomas Price, Nina Deliu, Audrey Durand, Michael Liut, Joseph Jay Williams
  Platform-based Adaptive Experimental Research in Education: Lessons Learned from The Digital Learning Challenge
  Proceedings of the 15th International Learning Analytics and Knowledge (LAK) Conference, 2025
  dl.acm.org/doi/full/10.1145/3706468.3706471
- **Harsh Kumar**, Ilya Musabirov, Mohi Reza, Jiakai Shi, Joseph J. Williams, Anastasia Kuzminykh, Michael Liut
  Guiding Students in Using LLMs in Supported Learning Environments: Effects on Interaction Dynamics, Learner Performance, Confidence, and Trust
  Proceedings of the ACM on Human-Computer Interaction (PACMHCI), CSCW, 2024 doi:10.1145/3687038
- Jessica Bo*, **Harsh Kumar***, Michael Liut, Ashton Anderson
  Disclosures & Disclaimers: Investigating the Impact of Transparency Disclosures and Reliability Disclaimers on Learner–LLM Interactions
  Proceedings of the Twelfth AAAI Conference on Human Computation and Crowdsourcing (HCOMP), 2024
  ojs.aaai.org/index.php/HCOMP/article/view/31597/33763
- **Harsh Kumar**, Ruiwei Xiao, Benjamin Lawson, Ilya Musabirov, Jiakai Shi, Xinyuan Wang, Huayin Luo, Joseph J. Williams, Anna Rafferty, John Stamper, Michael Liut
  Supporting Self-Reflection at Scale with Large Language Models: Insights from Randomized Field Experiments in Classrooms
  Proceedings of the Eleventh ACM Conference on Learning at Scale (L@S), 2024 doi:10.1145/3657604.3662042
- **Harsh Kumar**, Tong Li, Jiakai Shi, Ilya Musabirov, Rachel Kornfield, Jonah Meyerhoff, Ananya Bhattacharjee, Chris Karr, David Mohr, Anna Rafferty, Sofia Villar, Nina Deliu, Joseph Jay Williams
  Using Adaptive Bandit Experiments to Increase and Investigate Engagement in Digital Mental Health
  Proceedings of the Thirty-Sixth Annual Conference on Innovative Applications of Artificial Intelligence (IAAI-24), AAAI, 2024 ojs.aaai.org/index.php/AAAI/article/view/30328
- Ananya Bhattacharjee, Joseph J. Williams, Jonah Meyerhoff, **Harsh Kumar**, Alex Mariakakis, Rachel Kornfield
  Investigating the Role of Context in the Delivery of Text Messages for Supporting Psychological Wellbeing **(Best Paper Award, Top 1% of Submissions)**
  ACM Conference on Human Factors in Computing Systems (CHI), 2023
  dl.acm.org/doi/full/10.1145/3544548.3580774

- Rachel Kornfield, Caitlin A. Stamatis, Ananya Bhattacharjee, Bei Pang, Theresa Nguyen, Joseph J. Williams, **Harsh Kumar**, Sarah Popowski, Miranda Beltzer, Christopher J. Karr, Madhu Reddy, David C. Mohr, Jonah Meyerhoff
  A text messaging intervention to support the mental health of young adults: User engagement and feedback from a field trial of an intervention prototype
  Internet Interventions, 2023 sciencedirect.com/science/article/pii/S2214782923000672

- Jonah Meyerhoff, Miranda Beltzer, Sarah Popowski, Chris J. Karr, Theresa Nguyen, Joseph J. Williams, Charles J. Krause, **Harsh Kumar**, Ananya Bhattacharjee, David C. Mohr, Rachel Kornfield
  Small Steps over time: A longitudinal usability test of an automated interactive text messaging intervention to support self-management of depression and anxiety symptoms
  Journal of Affective Disorders, 2023 sciencedirect.com/science/article/pii/S0165032723013083

Working Papers

- **Harsh Kumar**, Jace Mu, Jonathan Vincentius, Ashton Anderson
  Beyond One-on-One: Human Learning in Multi-Agent LLM Environments
  (Under review at Learning at Scale, 2026)

- **Harsh Kumar**, Christopher M. Gideon, Nick DeFalco, Alyssa Rizzolo, Adam Coleman, Shamsi T. Iqbal
  From Metrics to Decisions: Designing LLM Support for Managerial Sensemaking About Organizational AI Adoption
  (work done during an internship at Microsoft)

- Sofia Panasiuk, Dana Kulzhabayeva, Mohi Reza, **Harsh Kumar**, Ananya Bhattacharjee, Felix Cheung, Joseph Jay Williams

  Doing Team Science at Scale: Practical Guidelines for Leading a Crowdsourced Scientific Collaboration in Psychology
  (Preprint: OSF. Under Review at Advances in Methods and Practices in Psychological Science (AMPPS).)

- **Harsh Kumar**, Ilya Musabirov, Jiakai Shi, Adele Lauzon, Kwan Kiu Choy, Ofek Gross, Dana Kulzhabayeva, Joseph J. Williams

  Exploring The Design of Prompts For Applying GPT-3 based Chatbots: A Mental Wellbeing Case Study on Mechanical Turk arXiv preprint arXiv:2209.11344 (Presented at CODE@MIT 2022.)

- Tong Li, Jacob Nogas, Haochen Song, **Harsh Kumar**, Audrey Durand, Anna Rafferty, Nina Deliu, Sofia S. Villar, Joseph J. Williams

  Algorithms for Adaptive Experiments that Trade-off Statistical Analysis with Reward: Combining Uniform Random Assignment and Reward Maximization
  arXiv preprint arXiv:2112.08507

Non-archival Publications

- **Harsh Kumar**, Mohi Reza, Jeb Thomas-Mitchell, Ilya Musabirov, Lisa Zhang, Michael Liut
  Understanding Help-Seeking Behavior of Students Using LLMs vs. Web Search for Writing SQL Queries
  DataEd '25: Proceedings of the 4th International Workshop on Data Systems Education, pp. 23–28, 2025
  doi:10.1145/3735091.373756

- Ruiwei Xiao, Xinying Hou, **Harsh Kumar**, Steven Moore, John Stamper, Michael Liut
  A Preliminary Analysis of Students' Help Requests with an LLM-powered Chatbot when Completing CS1 Assignments
  CSEDM'24: 8th Educational Data Mining in Computer Science Education Workshop, 2024

- **Harsh Kumar**, Yiyi Wang, Jiakai Shi, Norman Farb, Joseph J. Williams

  Exploring the Use of Large Language Models for Improving the Awareness of Mindfulness
  (Extended Abstract) CHI Conference on Human Factors in Computing Systems (CHI 2023)

- **Harsh Kumar**, Jiakai Shi, Ilya Musabirov, Adele Lauzon, Meagan Peters, Ofek Gross, Joseph J. Williams

  Prompt Engineering to Improve Experiences with Large Language Model-Administered Support: A Digital Experiment with Online Crowdworkers
  (Parallel Talk) Conference on Digital Experimentation (CODE 2022)

- **Harsh Kumar**, Kunzhi Yu, Andrew Chung, Jiakai Shi, Joseph J. Williams

  Exploring The Potential of Chatbots to Provide Mental Well-being Support for Computer Science Students
  (Poster) The ACM Technical Symposium on Computer Science Education (SIGCSE TS 2023)

- **Harsh Kumar***, Taneea S Agrawaal*, Kwan Kiu Choy, Jiakai Shi, Joseph J. Williams

  "Sounds like a Cheesy Radio Ad": Using User Perspectives for Enhancing Digital COVID Vaccine Communication Strategies for Public Health Agencies
  (Extended Abstract) CHI Conference on Human Factors in Computing Systems (CHI 2022)

- **Harsh Kumar**, Dana Kulzhabayeva, Joseph J. Williams

  Using Adaptive Experiments for Personalizing Mental-Health Related Interventions to Account for Heterogeneity between Participants
  (Poster) Intervention Science Preconference at Society For Personality and Social Psychology (SPSP 2023)

- Aayush Kapur, Himanshu Thakur, **Harsh Kumar**

  Safeguarding People against Social Media Frauds during the COVID-19 Oxygen Supply Crisis in India
  (Poster) NeurIPS Workshop on Machine Learning for the Developing World (ML4D, NeurIPS 2021)

Competitions and Awards

XPRIZE Digital Learning Challenge April 2023
University of Toronto *Toronto, ON*
Link to announcement

- The Adaptive Experimentation Accelerator team won the US$500,00 grand prize in the XPRIZE Digital Learning Challenge, a global competition to modernize, accelerate and improve the identification of effective learning tools and processes.
- I led the development of the Personalization System (MOOClet), incorporating contextual bandits, to allow researchers/instructors to conduct large-scale adaptive experiments at scale in classrooms.

DARPA AI Tools for Adult Learning (for my project "QuickTA") September 2023
University of Toronto *Toronto, ON*
Link to announcement

- QuickTA is an LLM-based tool to support students in the classroom or for self-directed learning. In contrast to many emerging tools, relying solely on LLM mechanisms for finding knowledge and ensuring retrieval, QuickTA brings together high quality educational content, Reinforcement Learning adaptive algorithms, and Intelligent Tutoring Systems to ensure that each student gets just-in-time support on the level they need: from hint to reflection prompt, from activating connection to previous knowledge to motivational support. In this project, TutorGen Inc works on bringing University of Toronto research behind QuickTA and Carnegie Mellon University expertise in AI systems for education to thousands of learners on Open Learning Initiative and other platforms.
- Through our initial rounds of deployments, we have provided LLM-enhanced support to more than 1,500 students learning how to write code, with more planned deployments for Statistics and Reading courses.

Teaching and Workshop Experience

Teaching Assistant September 2021 – Present
University of Toronto *Toronto, ON*

Courses: CSC428: Human-Computer Interaction, CSC2558: Multidisciplinary HCI, CSC318: Design of Interactive Media, CSC108: Introduction to Computer Programming, CSC343: Introduction to Databases, CSC2552: Computational Social Science

- Taught one-hour classes every week
- Assisted professors with course duties such as grading and handling LMS processes
- Facilitated discussions around in-class exercises
- Supported students in debugging their solutions

Session Chair (Track: AI and HCI) July 2024
10th International Conference on Computational Social Science (IC2S2) *Philadelphia, United States*

Co-organizer April 2023
CHI Conference on Human Factors in Computing Systems (CHI 2023) *Hamburg, Germany*

Workshop: Integrating Individual and Social Contexts into Self-Reflection Technologies

- The goal was to bring together a community of multidisciplinary researchers and practitioners who are seeking to design and develop reflection interventions that are situated within the fabric of the individual and social contexts of users.
- https://sites.google.com/view/reflectionchiworkshop2023/home

Expert Reviewer January 2024, January 2025
Tools Competition *https://tools-competition.org/*

Multi-Million Dollar Funding Opportunity for EdTech Innovation

- Invited to participate as an expert reviewer for the competition and provide feedback and guidance to qualified teams at different points of their journey.

References

- Dr. Ashton Anderson, Assistant Professor, Computer Science (CS), University of Toronto
`ashton@cs.toronto.edu`
- Dr. Jake M. Hofman, Senior Principal Researcher, Microsoft Research, New York City
`jmh@microsoft.com`
- Dr. Daniel G. Goldstein, Partner Research Manager, Microsoft Research, New York City
`dgg@microsoft.com`
- Dr. David M. Rothschild, Economist, Microsoft Research, New York City
`david@researchdmr.com`
- Dr. Shamsi Iqbal, Senior Principal Research Manager, Microsoft, Redmond
`shamsi@microsoft.com`
- Dr. Louis Tay, William C. Byham Professor, Department of Psychological Sciences, Purdue University
`stay@purdue.edu`
- Dr. Anastasia Kuzminykh, Assistant Professor, School of Information, University of Toronto
`anastasia.kuzminykh@utoronto.ca`
- Dr. Michael Liut, Assistant Professor (Teaching Stream), Department of Mathematical and Computational Sciences, University of Toronto Mississauga
`michael.liut@utoronto.ca`
- Dr. Rachel Kornfield, Research Assistant Professor, Preventive Medicine, Feinberg School of Medicine, Northwestern University
`rachel.kornfield@northwestern.edu`